

KRISTINE STANLEY

AI / ML ENGINEER

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PROFESSIONAL SUMMARY

Senior AI / ML Engineer with 11+ years across cybersecurity, connected mobility, insurance, and learning platforms, delivering Python 3.9-3.12, scikit-learn, PyTorch, FastAPI, MLflow, feature pipelines, cloud ML, PostgreSQL 12-16, Kubernetes solutions with secure APIs, scalable architecture, cloud delivery, observability, and production modernization that improves reliability, performance, release quality, and business scalability for enterprise products.

SKILLS

- **AI / ML Engineering:** Python 3.9-3.12, scikit-learn, PyTorch, FastAPI, MLflow, feature pipelines, cloud ML, PostgreSQL, Redis, Kafka, AWS, Azure, Docker, Kubernetes, observability, embeddings, evaluation, model serving, prompt quality, feature pipelines, retrieval systems, guardrails
- **AI Operations:** RAG evaluation, model monitoring, MLflow, vector databases, LangSmith, drift checks, human review workflows, secure data handling
- **Architecture:** clean architecture, domain-driven design, microservices, event-driven architecture, modular monoliths, CQRS, transactional outbox, API versioning
- **Cloud & DevOps:** Azure, AWS, Docker, Kubernetes, Terraform, Helm, GitHub Actions, Azure DevOps, Jenkins, CI/CD, feature flags, blue-green deployments
- **Databases & Messaging:** PostgreSQL, SQL Server, Redis, MySQL, MongoDB, Kafka, RabbitMQ, Azure Service Bus, AWS SQS, indexing, migrations, query optimization, caching
- **Testing & Quality:** xUnit, NUnit, Playwright, Cypress, Jest, Testcontainers, Pact, integration testing, contract testing, load testing, accessibility testing, static analysis, secure code review
- **Observability & Leadership:** OpenTelemetry, Serilog, Datadog, Application Insights, Prometheus, Grafana, incident response, root-cause analysis, mentoring, architecture reviews, stakeholder communication

RELEVANT EXPERIENCE

Senior Software Engineer

April 2021 - May 2026

Veracode Systems Inc — Remote / Full time, Somerville, MA

- Led senior delivery for enterprise application-security products using **Python 3.9-3.12, scikit-learn, PyTorch, FastAPI, MLflow, feature pipelines, cloud ML, PostgreSQL, Redis, Kafka, Docker, Kubernetes, OpenTelemetry, CI/CD** across AI-enabled service workflows, retrieval pipelines, evaluation dashboards, and secure automation tools for security, product, platform, and support teams.
- Reorganized tightly coupled services into domain-focused modules using **clean architecture**, typed contracts, dependency injection, and reviewable service boundaries, reducing recurring production defects by **36%**.
- Resolved P95 latency caused by slow retrieval queries, noisy context windows, model timeout handling, and expensive background jobs through **query tuning**, caching, pagination, asynchronous processing, and payload shaping, improving critical workflow response time by **43%**.
- Modernized legacy components from older runtime patterns to **AI / ML** practices, replacing obsolete packages, synchronous calls, unsafe shared state, and fragile deployment assumptions with maintainable release paths.
- Designed secure multi-tenant access patterns with **OAuth 2.0, OpenID Connect**, JWT validation, tenant-scoped data access, RBAC, audit logging, and policy-based authorization for sensitive customer data.
- Eliminated duplicate processing during retries by implementing **idempotency keys**, transactional state changes, outbox publishing, dead-letter replay procedures, and correlation identifiers across service boundaries.
- Built retrieval workflows, prompt governance, model routing, evaluation dashboards, and secure AI-assisted operations while coordinating backlog priorities, technical design, rollout planning, and dependency management with product managers, QA engineers, security reviewers, and customer-facing teams.
- Established automated quality gates with **evaluation suites, prompt regression checks, contract tests, offline replay tests, and human review sampling**, catching contract mismatches, migration failures, data regressions, and release-blocking defects before production deployment.

- Introduced operational dashboards with **OpenTelemetry, Serilog, Datadog, Application Insights, Prometheus, Grafana**, accelerating diagnosis of memory leaks, queue backlogs, database contention, failed integrations, and downstream timeout incidents.

Software Engineer

January 2019 - April 2021

Cubic Telecom USA — Remote / Full time, San Diego, CA

- Developed connected-mobility platform services using **Python 3.7-3.9, NLP services, Flask/FastAPI, PostgreSQL, RabbitMQ, Azure DevOps, model APIs** for device registration, SIM provisioning, subscription changes, carrier synchronization, billing events, and customer operations.
- Improved provisioning throughput by **38%** through asynchronous command handling, bounded concurrency, batched persistence, rate limiting, and resilient retry policies that protected external telecom APIs.
- Resolved failed activations caused by inconsistent carrier identifiers, incompatible payload structures, and delayed callbacks by creating canonical models, provider adapters, and replayable reconciliation workflows.
- Designed versioned API contracts with **OpenAPI**, validation rules, standardized error responses, permission-aware endpoints, and documented integration examples for portals and operational tools.
- Implemented message-driven workflows with durable queues, dead-letter handling, consumer checkpoints, idempotent updates, and correlation identifiers to prevent duplicate activations and preserve traceability.
- Corrected stale device states caused by out-of-order events through timestamp validation, monotonic transition rules, scheduled reconciliation jobs, and immutable status histories for support teams.
- Optimized slow device, account, and usage queries using covering indexes, compiled projections, connection-pool tuning, and removal of N+1 access patterns as connected-device volumes increased.
- Standardized delivery with **Docker, Azure DevOps**, automated tests, security scans, migration validation, environment checks, approval gates, and rollback notes for distributed releases.

Software Developer

January 2017 - December 2018

Asurion — Remote / Full time, Nashville, TN

- Built insurance and service-contract workflows using **Python 3.5-3.7, SQL Server, feature extraction scripts, REST integrations, reporting automation** for policy eligibility, claim intake, payment callbacks, appointment scheduling, repair status, and customer communications.
- Increased successful online claim completion by **27%** after redesigning orchestration flows to preserve recoverable state, return actionable validation details, and reduce unnecessary calls to slow dependencies.
- Created a backend-for-frontend and integration layer that combined customer, policy, inventory, payment, and appointment data while shielding product experiences from incompatible downstream contracts.
- Corrected duplicate appointment and payment records caused by repeated submissions through idempotency tokens, uniqueness constraints, transactional validation, and request-state verification.
- Implemented caching for frequently requested eligibility, location, and reference data with explicit invalidation, short expiration periods, and monitoring to improve speed without serving stale decisions.
- Developed scheduled reconciliation services for incomplete claims and delayed provider callbacks, recovering transactions after timeout conditions while preserving audit records for service and finance teams.
- Expanded automated coverage with **NUnit, Moq**, Selenium, service virtualization, API integration tests, and failure-scenario suites for claims, payments, appointments, and eligibility decisions.

Junior Software Developer

June 2015 - December 2016

Rustici Software — Hybrid / Full time, Nashville, TN

- Maintained learning-platform applications using **Python 3.4-3.5 scripts, SQL, MySQL, rule-based automation, C# service integrations** for course imports, learner-state persistence, launch configuration, completion tracking, reporting extracts, and xAPI-style event handling.
- Implemented administration workflows and integration endpoints for content ingestion, adding validation for malformed archives, incomplete manifests, missing metadata, and nonstandard customer payloads.
- Resolved excessive memory consumption during large imports by streaming compressed packages, limiting parallel extraction, disposing temporary resources, and recording actionable processing failures for support teams.
- Built defensive processing for duplicate and out-of-order learner events, preserving accurate completion, score, and resume state across platforms with inconsistent callback and retry behavior.
- Improved database reliability through targeted indexes, transactional updates, parameterized queries, and repeatable migration scripts for learner-state tables used during high-volume reporting periods.
- Collaborated with senior engineers and support specialists to reproduce unusual integration failures, document payload assumptions, and deliver maintainable fixes that shortened customer troubleshooting cycles.

SELECTED PROJECTS

- **Production LLM Workflow Platform:** Delivered a production AI workflow with **Python 3.9-3.12, scikit-learn, PyTorch, FastAPI, MLflow, feature pipelines, cloud ML, PostgreSQL, Redis, Kafka, AWS, Azure, Docker, Kubernetes, observability**, retrieval controls, evaluation datasets, human review queues, and secure API integration for internal support and engineering teams.
- **AI Evaluation and Observability Framework:** Created a model quality framework with regression prompts, offline replay, latency tracking, cost dashboards, drift checks, and alerting for degraded answer quality or unsafe outputs.

EDUCATION

Tennessee State University — Bachelor's Degree in
Computer Science

Sep 2011 - May 2015

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate
- Microsoft Certified: Azure Solutions Architect Expert
- AWS Certified Developer - Associate
- Professional Scrum Master I - PSM I